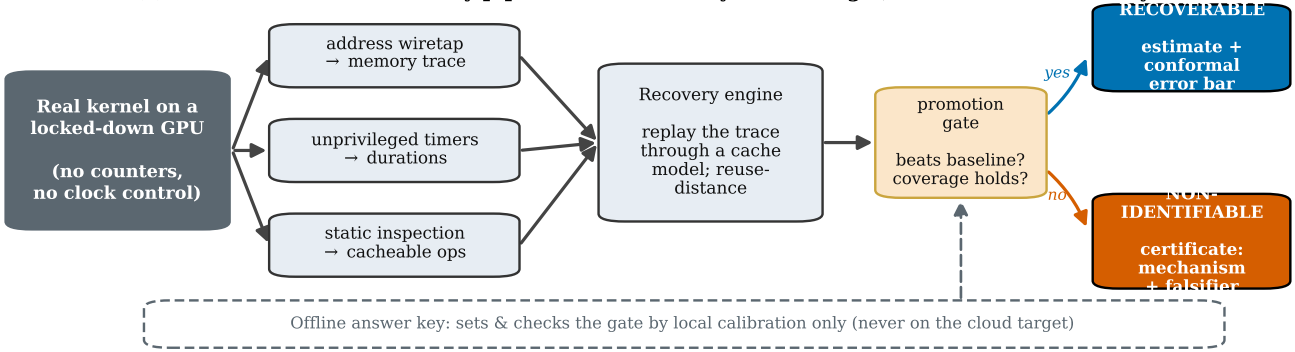
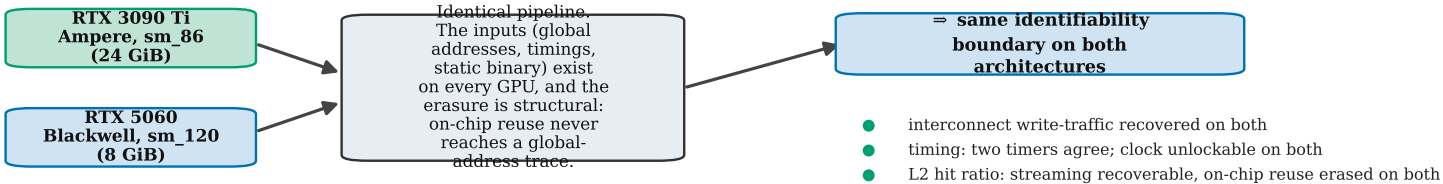
(a) The counter-free recovery pipeline: observe only what is legal, then recover or certify



(b) Why the result transfers across architectures: the inputs are architecture-agnostic and the erasure is structural



Per-architecture calibration differs (e.g. read-traffic recovered on Blackwell, declined on Ampere); the boundary does not.